


THE UNIVERSITY OF MELBOURNE

# FNCE 30007 Derivative Securities

## Lecture 2– Hedging with Futures

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# Outline

- **Short/Long Hedges**
- Basis Risk
  - Cross Hedging
- Use of Stock Index Futures
- Hedge Ratio

**Reading:** Chapter 3

# Hedging

- Futures can be used by speculators to gain highly levered, risky positions (lecture 1)
- This lecture focuses on the use of futures by hedgers
- Hedger's objective: take a futures position to reduce risk.

# Short and Long Hedges

Hedgers aim to use futures markets or forward contracts to reduce a particular risk they may face.

A short hedge is a hedge which involves a short position in futures contracts.

- It is appropriate when the hedger already owns an asset (or will own it at some definite date) and expects to sell it at some time in the future.
- This allows them to lock in the price they will receive.

A long hedge involves taking a long position in futures contracts.

- It is appropriate when a company knows it will have to purchase a certain asset in the future and wants to lock in the price now.

# Short hedge

- April 20: Farmer negotiates to sell 50,000 bu of corn at the spot price on June 20
- June 20 is futures maturity date – no basis risk ($S_T=F_T$)
- Quoted:
  - Spot price of corn: \$3.50/bu
  - June corn futures price: \$3.35/bu (each contract 5,000 bu)




- After futures gains & losses, price received is \$3.35/bu.

# Short hedge

## Scenario 1: Spot price June 20 $3.10/bu

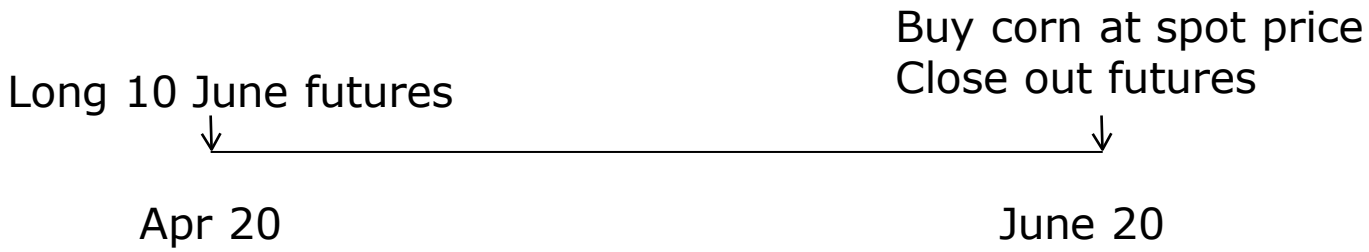
Spot: Farmer sells corn at ($3.10)(50,000)= 155,000
Futures: Gain ($3.35 – $3.10)(50,000) = $12,500
Total received: $155,000 + $12,500 = $167,500.
price/bu = $167,500 / 50,000 = $3.35
Alternatively price/bu = $3.10 + ($3.35 – $3.10) = $3.35

## Scenario 2: Spot price June 20 $3.70/bu

Spot: Sell corn at ($3.70)(50,000) = $185,000
Futures: Loss ($3.70 – $3.35)(50,000) = $17,500
Total received: $185,000 – $17,500 = $167,500.
Price/bu = $167,500 / 50,000 = $3.35
Alternatively price/bu = $3.70 - ($3.70 – $3.35) = $3.35

# Long hedge

- April 20: Food company to buy 50,000 bu of corn at the spot price on June 20
- June 20 is futures maturity date – no basis risk ($S_T=F_T$)
- Quotes:
  - Spot price of corn: $3.50/bu
  - June corn futures price: $3.35/bu (each contract 5,000 bu)




- After futures gains & losses the price paid should be $3.35/bu.

# Long hedge

## Scenario 1: Spot price June 20 $3.10/bu

Spot: Company buys corn ($3.10)(50,000) = 155,000

Futures: Loss ($3.35 – $3.10)(50,000) = $12,500

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# Basis Risk

A perfect hedge is one that completely eliminates the risk. **However, a perfect hedge is very rare.**

Some of the reasons for this are:

- The asset whose price is to be hedged may not be exactly the same as the asset underlying the futures contract; e.g. An airline company uses heating oil futures contracts to hedge its exposure to fluctuations in aviation fuel prices
- The hedge may require the futures contract to be closed out before its delivery month.
- The size of the contract may not exactly cover your position.

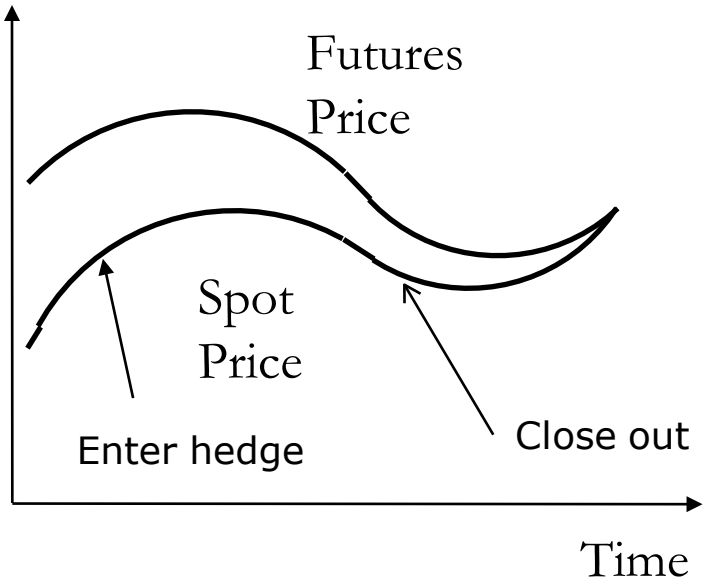
# Basis Risk

What is basis risk:

• If the asset to be hedged and the asset underlying the futures contract are the same, the basis risk should be zero at the expiration of the futures contract.

• The formula for working out the basis in a hedging situation is:

$$\text{Basis} = \text{Spot price of asset to be hedged} - \text{Futures price of contract used}$$




Convergence explained by the cost of carry model (next week)

# Basis risk

- Prior to expiration, the basis may be positive or negative.
- When the spot price increases by more than the futures price, the basis increases. We call this strengthening of the basis.
- When the futures price increases by more than the spot price, the basis declines. We call this weakening of the basis.

# Basis risk: Long hedge

- Suppose

  $F_0$: Initial Futures Price

  $F_1$: Final Futures Price

  $S_1$: Final Asset Price

  } Unknown at time t=0

- Hedge future asset purchase by taking a long futures position
- Cost of Asset = $S_1 - (F_1 - F_0) = F_0 + (S_1 - F_1) = F_0 + \text{Basis}_1$
- For a perfect hedge, $S_T = F_T$, so basis =0

# Basis risk: Short hedge

- Suppose

  $F_0$ : Initial Futures Price

  $F_1$ : Final Futures Price

  $S_1$ : Final Asset Price

- Hedge the future asset sale via a short futures contract
- Price Realised=$S_1$+ ($F_0$ – $F_1$) = $F_0$ + ($S_1$ - $F_1$)= $F_0$ + $\text{Basis}_1$
- For a perfect hedge, $S_T$=$F_T$, so basis =0

# Basis Risk

One key factor affecting basis risk is the choice of the futures contract to be used for hedging. This choice has the following components:

- When there is no futures contract on the asset being hedged, choose the contract whose futures price is most highly correlated with the asset price.
- Choose a delivery month that is as close as possible to, but later than, the end of the life of the hedge.
- When the exposure cannot be matched exactly by an integer number of futures contracts, use slightly more or fewer contracts, depending on the hedger's objectives, risk tolerance, and market view.

# Basis risk: cross hedge

Cross hedging occurs when the asset underlying the futures contract is different to the asset whose price is being hedged.

- For Example: an airline company may be concerned about the future price of aviation fuel. However, as there are no futures contracts on aviation fuel, the company chooses to use heating oil futures contracts to hedge its exposure.

# Basis risk: cross hedge

- With cross hedge, basis risk likely to increase.

Additional source of uncertainty

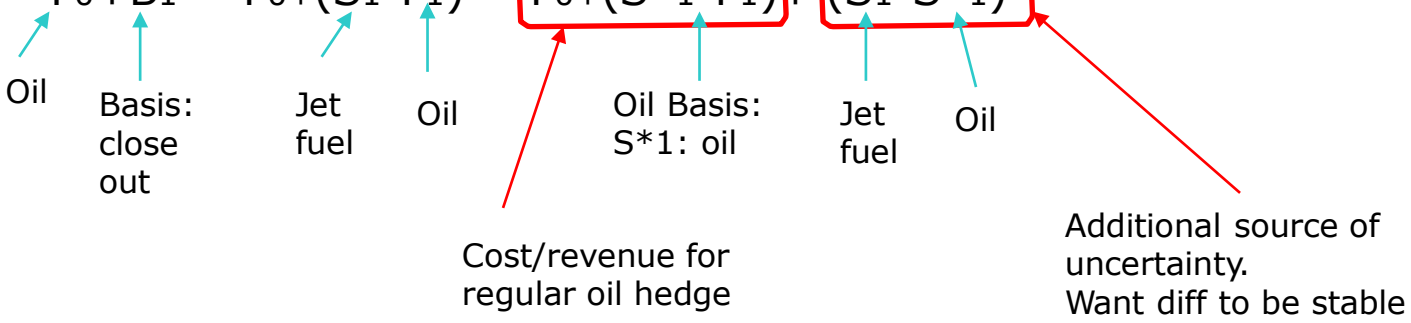
$$F_0 + (S^*_1 - F_1) + (S_1 - S^*_1)$$

$(S^*_1 - F_1)$ = basis that exists if the asset being hedged were the same as the asset underlying the futures

$(S_1 - S^*_1)$ = basis from difference between two assets.

# Basis risk: cross hedge

- Illustrate with a jet fuel hedge using oil futures
- $F_0 + B_1 = F_0 + (S_1 - F_1) = F_0 + (S^*_1 - F_1) + (S_1 - S^*_1)$



# Outline

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- **Use of Stock Index Futures**
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# Hedging Using Index Futures

Stock index futures can be used to hedge an equity portfolio.

To hedge the risk in a portfolio, the number of contracts that should be shorted is:

$$N^* = \beta \frac{V_A}{V_F}$$

Where $V_A$ is the value of the portfolio, $\beta$ is its beta (from CAPM), and $V_F$ is the value of the assets underlying one futures contract.

# Hedging Using Index Futures

Example:

- Imagine that the value of SPI200 is 3500.
- The value of the portfolio to be hedged is $5 million.
- The beta of the portfolio is 1.5.

What position in SPI200 futures contracts is necessary to hedge the portfolio?

# Hedging Using Index Futures

$$\beta \frac{V_A}{V_F} = 1.5 \text{ x } 5\text{m}/3500\text{x}25 = 86 \text{ contracts (SHORT)}$$

- What position is necessary to reduce the beta of the portfolio to 0.75?

# Hedging Using Index Futures

What position is necessary to reduce the beta of the portfolio to 0.75 (β*)?

Given we were hedging 100% with 86 contracts to reduce our risk to zero, we can take 43 to hedge 50% to give us half of our previous risk of 1.5.

Generally:

$$(\beta - \beta*)\frac{V_A}{V_F} = (1.5 - .75)\frac{5m}{87500} = 43$$

What position is necessary to increase the beta of the portfolio to 2.00?

# Hedging Using Index Futures

What position is necessary to increase the beta of the portfolio to 2.00?

$$(\beta - \beta *)\frac{V_A}{V_F}$$

$$(1.5 - 2.0)\frac{5M}{3500x25} = -29$$

- Since this is negative we must go LONG!!!

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# Minimum variance hedge ratio (MVHR)

$$\text{Hedge ratio (HR)} = \frac{\text{size of futures position}}{\text{size of spot exposure}}$$

- Typically measured in quantities (commodity contracts) and $ values (financial contracts)
- So far the hedge ratio = 1.0 (naïve hedge)
- A naïve hedge may be sub-optimal with basis risk i.e
  - cross hedging and/or
  - hedge completion date not same as futures expiration date

# Minimum variance hedge ratio (MVHR)

- We can derive a minimum variance hedge ratio
- It recognises basis risk and seeks to minimise the variation in the hedged profit/loss.

$$h^* = \frac{\sigma_{sf}}{\sigma_f^2} = \rho_{sf}\frac{\sigma_s}{\sigma_f}$$

$h^* = \text{optimal hedge ratio}$

$\sigma_{sf} = \text{covariance between the spot exposure and futures used}$

$\sigma_f^2 = \text{futures variance}$

$\sigma_s = \text{spot std deviation}$

$\rho_{sf} = \text{correlation between spot and futures}$

# Minimum variance hedge ratio (MVHR)

- Example 1: Hedge S&P500 (S&P500.xlsx) for index fund
  - Calculate variances, covariance and correlations using continuously compounded returns

$$r_{s,t} = 100 \times \left[\ln(S_t) - \ln(S_{t-1})\right]$$

  - Calculate the cont compounded hedged returns each day via

$$r_{h,t} = r_{s,t} - h^* \times r_{f,t}$$

  - Determining the time period for estimation
  - Comparison of naïve v MVHR
  - Can calculate the optimal number of contracts via

$$N^* = h^* \frac{V_A}{V_F}$$

$V_A$=\$10M, $V_F$=\$1M
Naïve hedge N*=10
If h*=0.9, N*=9

# Minimum variance hedge ratio (MVHR)

| Portfolio return summary measures from July 2024 to June 2025 | | | | | | | |
|---|---|---|---|---|---|---|---|
| Investment $ | 200,000,000 | | | | | | |
| | Naïve | Full DS | GFC | COVID | Last 3 years | Perfect foresight | Unhedged |
| Total return % | 0.2227 | 0.4062 | 0.3730 | 0.3219 | 0.3001 | 0.3616 | 9.1029 |
| Avg return per day % | 0.0009 | 0.0017 | 0.0015 | 0.0013 | 0.0012 | 0.0015 | 0.0375 |
| Std dev | 0.2243 | 0.2222 | 0.2224 | 0.2228 | 0.2231 | 0.2225 | 1.2615 |
| Sharpe ratio | 0.0041 | 0.0075 | 0.0069 | 0.0059 | 0.0055 | 0.0067 | 0.0297 |
| | | | | | | | |
| Portfolio value | 200,445,963 | 200,814,044 | 200,747,423 | 200,644,920 | 200,601,159 | 200,724,463 | 219,060,229 |
| | | | | | | | |
| 95 % Conf interval | | | | | | | |
| Daily returns | | | | | | | |
| Lower bound (%) | -0.439 | -0.434 | -0.434 | -0.435 | -0.436 | -0.435 | -2.435 |
| Upper bound (%) | 0.441 | 0.437 | 0.437 | 0.438 | 0.438 | 0.438 | 2.510 |
| | | | | | | | |
| $ investment | | | | | | | |
| Lower bound ($) | -$877,607 | -$867,865 | -$868,731 | -$870,840 | -$872,026 | -$869,121 | -$4,870,045 |
| Upper bound ($) | $881,273 | $874,551 | $874,871 | $876,139 | $876,967 | $875,073 | $5,019,888 |

# Minimum variance hedge ratio (MVHR)

- Example 2: Fuel hedge (Fuel hedge.xls)

$$N^* = \frac{h^* V_A}{V_F}$$

$V_A$ = size of position being hedged (units)
$V_F$ = size of one futures contract
N* = optimal number of futures contracts

# Next Week

Forward and Futures Pricing:

- Introduction
  - Investment v consumption assets
  - Measuring interest rates
- Investment assets
  - No income
  - Known income (bond futures)
  - Known yield (stock index futures)
- Consumption assets